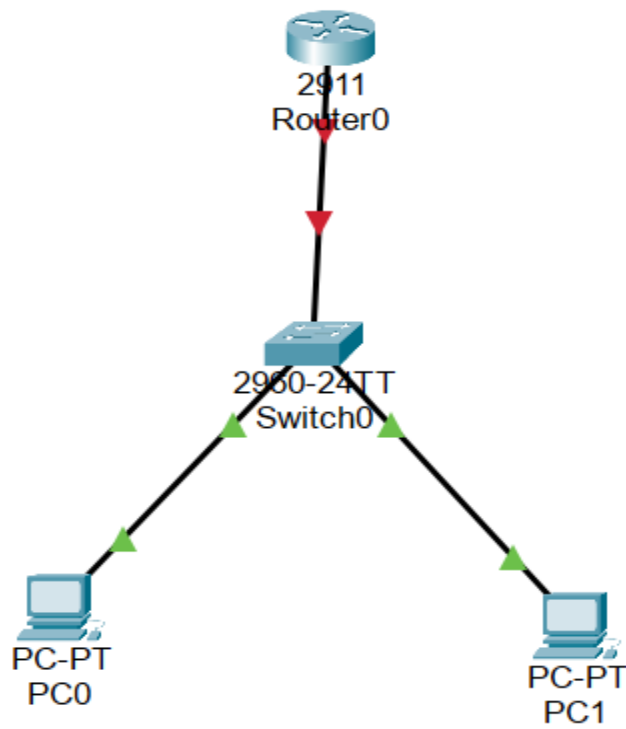


Enterprise Network Design & Documentation – Week 9

1) Network Topology Description:

The network consists of one router and one managed switch configured with multiple VLANs to simulate an enterprise environment. VLAN 10 represents the HR department and VLAN 20 represents the IT department. Inter-VLAN routing is implemented using Router-on-a-Stick configuration. The network topology is as follows:



2) VLAN Table

◆ Create a Table Like This

VLAN ID	VLAN Name	Department	IP Range
10	HR	Human Resources	192.168.10.0/24
20	IT	Information Technology	192.168.20.0/24

◆ Explanation Text

VLANs are used to logically segment the network into different departments. This improves security, reduces broadcast traffic, and simplifies network management.

3) IP Addressing Table

◆ Table Format

Device	Interface	IP Address	Subnet Mask	Gateway
Router	G0/0.10	192.168.10.1	255.255.255.0	—
Router	G0/0.20	192.168.20.1	255.255.255.0	—
PC-HR	NIC	192.168.10.10	255.255.255.0	192.168.10.1
PC-IT	NIC	192.168.20.10	255.255.255.0	192.168.20.1

◆ Explanation

Each VLAN has a dedicated IP subnet. The router sub-interfaces act as the default gateway for devices in each VLAN.

4) Devices Used

◆ Add This Table

Device	Model	Purpose
Router	Cisco 2911	Inter-VLAN routing
Switch	Cisco 2960	VLAN segmentation
PC	Generic PC	End-user simulation
Software	Cisco Packet Tracer	Network simulation

5) Issues Faced & Solutions

◆ Use This Format (Very Important)

✗ Issue 1: No communication between VLANs

Cause: Inter-VLAN routing not configured

Solution: Router sub-interfaces were created using 802.1Q encapsulation

Issue 2: PC not receiving IP address

Cause: Incorrect gateway configuration

Solution: Correct default gateway was assigned

Issue 3: Trunk not working

Cause: Switch port not set to trunk mode

Solution: Configured port using switchport mode trunk

6) Conclusion

This documentation provides a structured overview of the enterprise network configuration, ensuring easy troubleshooting, scalability, and maintainability.

DAY 67 – Weekly Quiz & Review

Short Summary

Write This

In Week 9, I learned core enterprise networking concepts such as VLAN implementation, inter-VLAN routing, Spanning Tree Protocol, DHCP relay, and Router-on-a-Stick configuration. These technologies are widely used in enterprise networks to improve performance, security, and scalability.

Concepts Review

◆ VLAN vs Subnet

VLAN	Subnet
Logical segmentation	IP-based segmentation
Works at Layer 2	Works at Layer 3
Improves security	Controls IP routing

Explanation:

VLANs separate devices logically on the same switch, while subnets separate devices at the network layer using IP addressing.

◆ **Purpose of STP**

Spanning Tree Protocol prevents network loops by blocking redundant paths, ensuring a loop-free topology in switched networks.

◆ **DHCP Relay**

DHCP Relay allows clients in one subnet to receive IP addresses from a DHCP server located in another subnet using the `ip helper-address` command.

◆ **Router-on-a-Stick**

Router-on-a-Stick enables inter-VLAN routing using a single router interface with multiple sub-interfaces configured for different VLANs.

3 What I Learned

- ✓ Enterprise VLAN design
- ✓ Inter-VLAN routing methods
- ✓ Importance of STP in redundancy
- ✓ DHCP services across subnets
- ✓ Professional network documentation

4 Challenges Faced

- Understanding VLAN tagging
- Configuring router sub-interfaces
- Trunk vs access ports confusion
- IP planning for multiple subnets

◆ How I Overcame

By practicing multiple labs in Packet Tracer, referring to Cisco documentation, and troubleshooting step-by-step.

5 Final Reflection

Week 9 significantly improved my understanding of enterprise-level networking and prepared me for real-world network administration tasks.
